

Alexander LaChapelle

github.com/ajlachapelle

Education

- **M.Eng. in Computer Science (Expected Fall 2026)**
 - **Oregon State University**
 - Coursework: Algorithms, Software Maintenance, Parallel Programming, Shaders, Quantum Computing, Machine Learning, Deep Learning, Foundation Models
- **B.Sc. in Mathematics and Computer Science**
 - **University of Oregon**
 - Computer Science coursework: C/C++ and Unix, Data Structures, Algorithms, Computer Organization (C/IA32), Computer Graphics (C/C++, OpenGL), Networks, Principles of Programming Languages
 - Mathematics coursework: Linear Algebra, Numerical Analysis, Discrete Mathematics, Several-Variable Calculus, Differential Equations, Proofs
- **Minor in Physics**
 - Physics coursework: Mechanics, Thermodynamics, Electrodynamics

Projects and Code Samples

- **Current Prediction GCN**
 - Designed a Graph Convolutional Neural Network to meet specifications for analog circuit representation
 - Implemented the network in Python
- **TopoMap Implementation**
 - Analyzed the TopoMap dimensionality reduction algorithm from an academic paper
 - Implemented the algorithm in Python
- **3D Geometry Renderer**
 - Wrote a scanline-rasterizer in C/C++

Skills

Languages: C/C++, Python, Java, C#

Operating Systems: Linux, Windows

Version Control: Git, Github

Development Tools: Visual Studio, Visual Studio Code, GDB, Docker

Interests: AI, scientific computing